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Exploring the Impact of Particle Size and Shape on Predicting Post-Blast Origin of Small Arms Propellant (SAP) Particles.

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Introduction:
Small arms propellants (SAP), commonly known as canister powders, serve a dual purpose within the realm of firearms enthusiasts. While these materials facilitate the legitimate assembly of ammunition, their accessibility and cost-effectiveness also render them attractive for the construction of improvised explosive devices (IEDs). The utilization of SAP particles in such illicit activities underscores the necessity for robust forensic analysis techniques to discern the post-blast origin of these particles. In response to this imperative, this project endeavors to employ advanced statistical methodologies to develop predictive models capable of differentiating between SAP particles recovered from exploded and unexploded IEDs.

Problem Statement:
The forensic examination of SAP particles recovered from incidents involving IEDs presents a formidable challenge to investigative agencies. Currently, there exists a critical need for predictive models that can accurately classify SAP particles based on their post-blast origin. Dr. Jack Hietpas of John Jay College of Criminal Justice has underscored the suspicion that particles retrieved from exploded IEDs may exhibit distinct size characteristics compared to those from unexploded devices. To address this concern and facilitate forensic investigations, this project aims to build predictive models leveraging quantitative image analysis data to discern the provenance of SAP particles.

Objectives:
1. Develop robust predictive models capable of classifying SAP particles as originating from exploded or unexploded IEDs, utilizing data acquired through high-throughput, non-destructive, and low-cost image analysis routines.
2. Investigate the validity of Dr. Hietpas' suspicion regarding the size characteristics of SAP particles recovered from exploded IEDs, by analyzing the predictive models' interpretations and insights derived from statistical analyses.
3. Explore the impact of various statistical techniques and feature engineering strategies on the performance of predictive models, with a focus on optimizing classification accuracy and interpretability.

Methodology
1. **Creation of Binary Variable:**
    - A new binary variable was created to encode the post-blast origin of SAP particles. Particles recovered from exploded IEDs were labeled as "exploded" (coded as 1), while those from unexploded devices were labeled as "unexploded" (coded as 0). This binary variable serves as the response variable for the predictive modeling task.
2. **Data Preparation and Training Set Creation:**

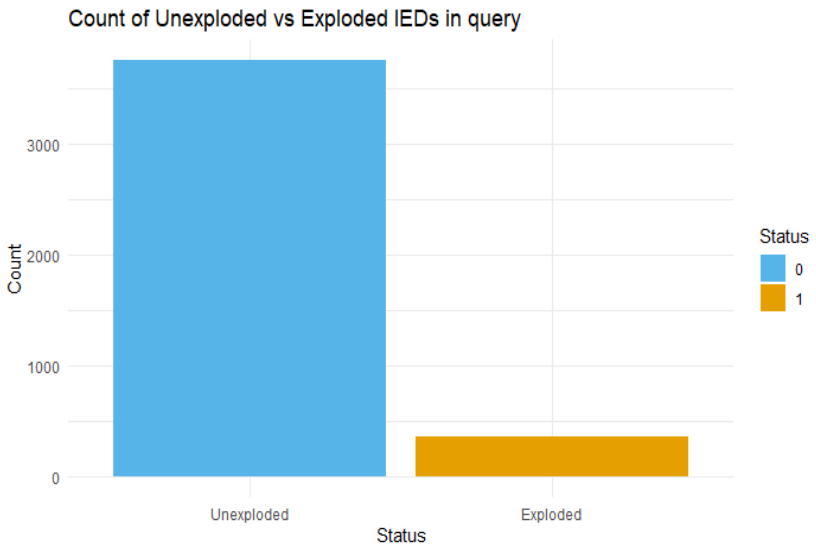
- The query datasets containing particles recovered from exploded and unexploded IEDs were merged into a single dataset to form the training dataset for model development.
- To ensure balanced representation of both classes in the training dataset, a class balance was performed. This involved randomly sampling an equal number of exploded and unexploded particles, thereby mitigating potential biases introduced by class imbalance. Balanced datasets improve the generalization performance of predictive models by preventing them from being skewed towards the majority class.

3. **Model Selection and Training:**
    - Logistic regression was chosen as the primary modeling technique due to its simplicity, interpretability, and suitability for binary classification tasks.
    - The logistic regression model was trained using 10-fold cross-validation, repeated three times to ensure robustness and reliability of model performance estimates.
    - Cross-validation involves partitioning the dataset into k subsets (folds), training the model on k-1 folds, and evaluating its performance on the remaining fold. This process is repeated k times, with each fold serving as the test set exactly once. Repeating this process multiple times helps in reducing the variability in performance estimates and provides more reliable evaluations of the model's generalization performance.
4. **Model Evaluation:**
    - The trained logistic regression model was evaluated using the merged dataset containing particles from both exploded and unexploded IEDs.
    - Performance metrics such as accuracy were calculated to assess the model's ability to correctly classify particles based on their post-blast origin.
5. **Interpretation and Analysis:**
    - The coefficients of the logistic regression model were examined to interpret the impact of predictor variables (particle size and shape features) on the likelihood of a particle being from an exploded or unexploded IED.
    - Insights derived from the model interpretation were used to validate Dr. Hietpas' suspicion regarding the size characteristics of SAP particles recovered from exploded IEDs.
    - Further analyses were conducted to explore the significance of individual features and their contributions to the predictive performance of the model.

Data Exploration

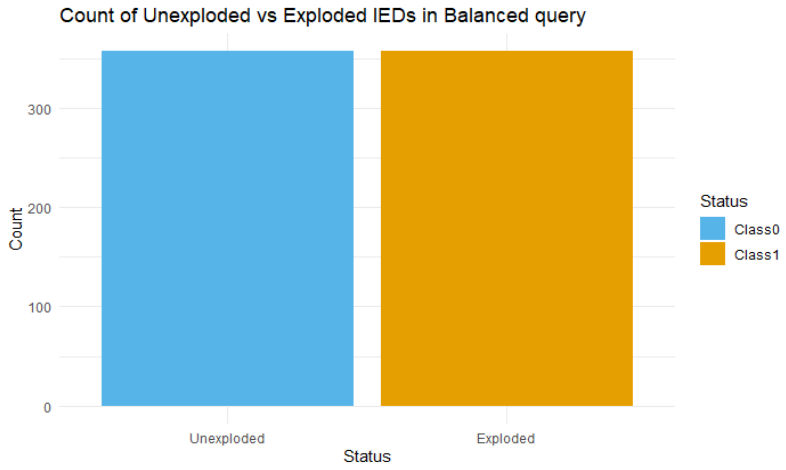
Initially, the training dataset that is our **query dataset (exploded and unexploded) 4109 observations**. As we can see in the figure attached.

unexploded	exploded
3752	357



To avoid class imbalance, which might lead to biased model predictions and decreased classification performance, we randomly selected an equal number of observations from each category **Query Balanced dataset (Exploded and Unexploded).** This ensures that both exploded and unexploded IEDs are represented proportionally in the training dataset, allowing the model to learn from a balanced distribution of classes and make more accurate predictions for both classes. As we can see in the figure its equally distributed.

unexploded	exploded
357	357

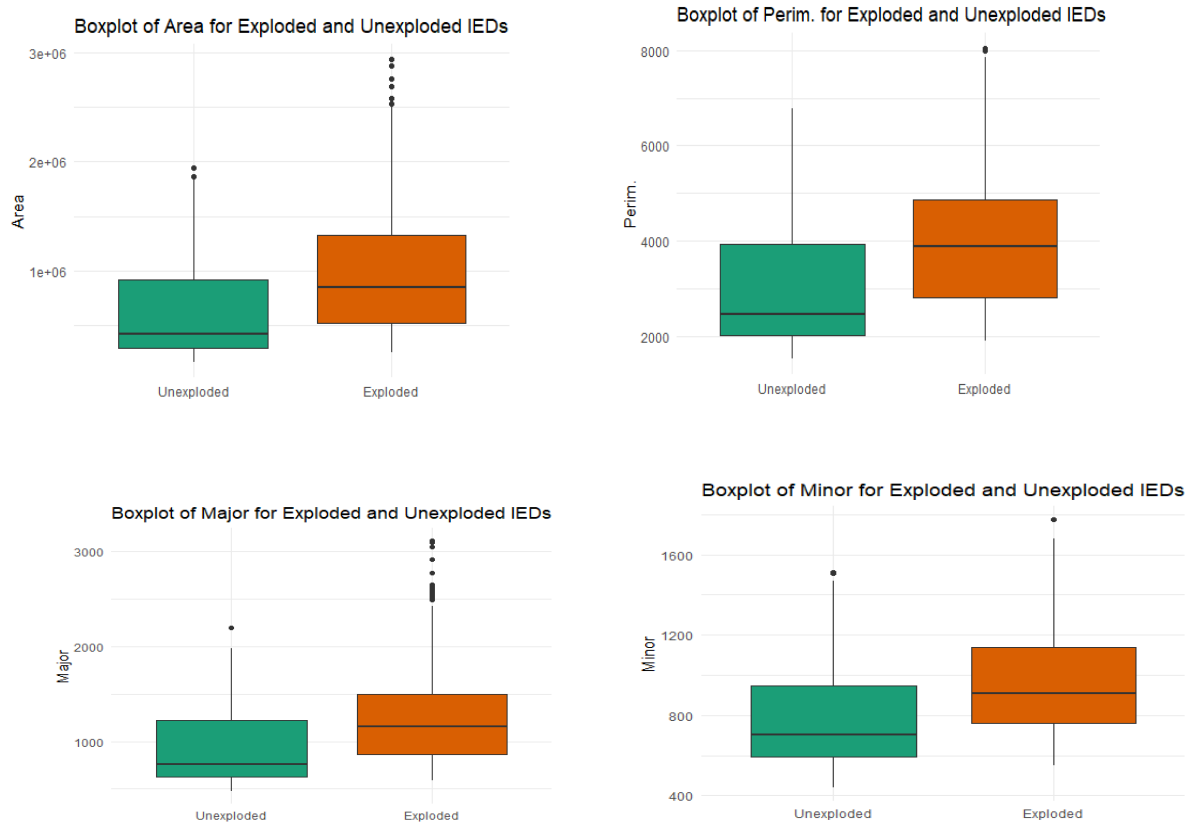


Variable Selection

The predictor variables utilized in the model were carefully chosen based on insights gleaned from past research. Specifically, the selected predictors offer a nuanced understanding of the size characteristics of particles recovered from IEDs. **These predictors include Area, Perimeter,**

Major, and Minor, as elucidated in a seminal publication (https://www.sciencedirect.com/science/article/pii/S0379073823003304?via%3Dihub). Each of these variables provides crucial information regarding the physical dimensions of the particles, thereby offering valuable insights into the composition and potential origin of the IEDs. By leveraging these predictor variables, the model endeavors to discern patterns and relationships that elucidate the post-blast origin of small arms propellant (SAP) particles with a heightened level of precision and accuracy.

Relationship Between Predictor Variables and Target Variable (Q



While observing the four plots displaying predictors against the target variable, a notable pattern emerges: the sizes of 'Exploded' instances appear higher than 'Unexploded'. However, it's essential to exercise caution before generalizing this observation as evidence of a definitive relationship. The apparent disparity in sizes could potentially be attributed to various factors beyond a direct causal link. Factors such as sampling bias, data collection methods, or inherent variability in the dataset may contribute to this pattern.

Logistic Model.

The logistic regression model was trained using 10-fold cross-validation with 3 repetitions on a dataset consisting of 714 samples and 4 predictor variables. The target variable had 2 classes: 'Class0' and 'Class1'. No pre-processing steps were applied to the predictor variables before model training. The cross-validated accuracy of the model was approximately 61.73%, indicating that it correctly classified around 61.73% of the instances. This metric provides valuable insight into the model's performance in predicting the outcome classes based on the provided predictor variables.

Testing Model

Coefficients:	**Estimate**	**Std. Error**	**z value**	**Pr(>|z|)**
(Intercept)	-5.12E+00	9.05E-01	-5.659	1.52e-08 ***
Area	-3.36E-06	1.08E-06	-3.102	0.001920 **
Perim.	3.74E-04	3.47E-04	1.08	0.280123
Major	2.25E-03	1.30E-03	1.738	0.082213 .
Minor	4.73E-03	1.27E-03	3.729	0.000192 ***

Interpretation

The logistic regression model coefficients provide insights into the relationship between the predictor variables (particle size features) and the probability of a particle being recovered from an exploded IED.

Intercept: The intercept represents the log-odds of the outcome (particle being from an exploded IED) when all predictor variables are zero. In this case, it suggests the baseline probability of a particle being from an exploded IED when all other factors are absent.

Area: The coefficient for "Area" indicates the change in the log-odds of the outcome for a one-unit increase in the area of the particle. A negative coefficient suggests that an increase in particle area is associated with a decrease in the log-odds of the particle being from an exploded IED.

Perimeter (Perim.): The coefficient for "Perimeter" represents the change in the log-odds of the outcome for a one-unit increase in the perimeter of the particle. However, with a p-value of 0.280123, this coefficient is not statistically significant.

Major: The coefficient for "Major" indicates the change in the log-odds of the outcome for a one-unit increase in the major axis of the particle's fitted ellipse. However, with a p-value of 0.082213, this coefficient is marginally significant and may not provide strong evidence of an association.

Minor: The coefficient for "Minor" represents the change in the log-odds of the outcome for a one-unit increase in the minor axis of the particle's fitted ellipse. A positive coefficient suggests that an increase in the minor axis is associated with an increase in the log-odds of the particle being from an exploded IED.

Regarding Dr. Hietpas' suspicion concerning the size of particles recovered from exploded IEDs:

- The negative coefficient for "Area" suggests that larger particles are less likely to be recovered from exploded IEDs, which aligns with the suspicion that smaller particles are more easily consumed in the blast.

- The positive coefficient for "Minor" supports the idea that particles with a larger minor axis are more likely to be from exploded IEDs, potentially indicating fragmentation or alteration of particle shape due to the explosion.

In summary, the coefficient interpretations partially support Dr. Hietpas' suspicion regarding the size of particles recovered from exploded IEDs, particularly with respect to particle area and minor axis. However, the interpretation should be cautious, especially for variables with marginal statistical significance. Additional analyses and validation may be warranted to further confirm the findings.

Merged Datset (exploded and Unexploded) is then fitted for the prediction, and we can see the confusion matrix below. **So, in simple words we use query dataset which has 357 observations for each exploded and unexploded and then to test for the prediction we use next dataset that is the Merged Dataset (Exploded and Unexploded).**

Figure: Merged Dataset Box plot

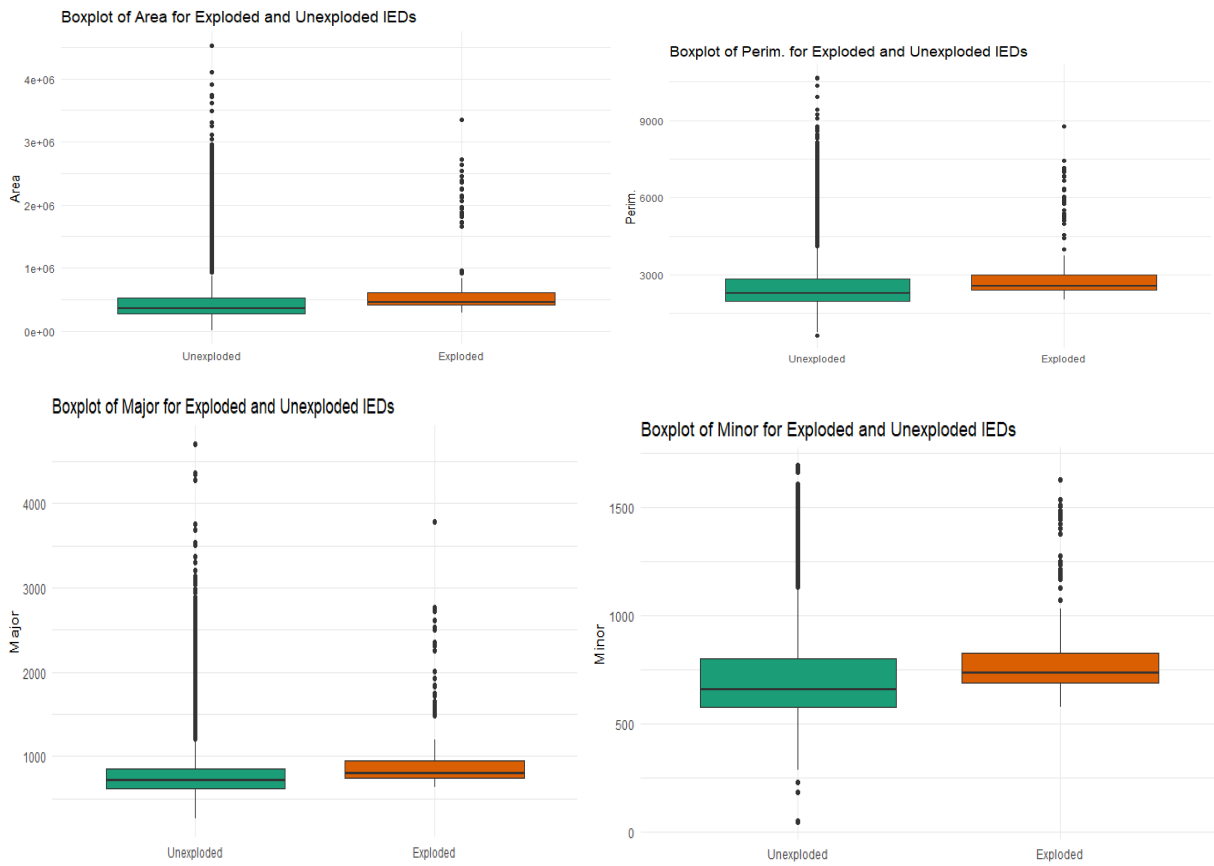


Table: No. of exploded and Unexploded in **Merged Dataset** (exploded and unexploded)

Unexploded	Exploded
9649	189

Confusion Matrix

Prediction	Reference	
	Unexploded	**Exploded**
Unexploded	7640	138
Exploded	2009	51

The accuracy of the predictive model, as indicated by the confusion matrix, is 0.7818 or 78.18%. This means that the model correctly classified approximately **78.18%** of the SAP particles into their respective classes (exploded or unexploded IEDs) based on the provided features.

Wilcoxon Sign Ranked Test

Wilcoxon rank sum test with continuity correction
data: Area by Status
W = 588416, p-value < 2.2e-16
alternative hypothesis: true location shift is less than 0

Wilcoxon rank sum test with continuity correction

data: Minor by Status
W = 605639, p-value = 1.203e-15
alternative hypothesis: true location shift is less than 0

The Wilcoxon rank sum test was conducted to assess the statistical significance of the difference in 'Area' and 'Minor' between particles recovered from exploded and unexploded IEDs. The alternative hypothesis for both tests posit that particles from exploded IEDs have lesser values of 'Area' and 'Minor' compared to particles from unexploded IEDs.
For the 'Area' variable, the Wilcoxon test yielded a p-value less than 2.2e-16, indicating a highly significant difference in 'Area' between the two groups. This suggests that particles recovered from exploded IEDs tend to have lesser 'Area' values compared to particles from unexploded IEDs.
Similarly, for the 'Minor' variable, the Wilcoxon test resulted in a p-value of 1.203e-15, indicating a statistically significant difference in 'Minor' between particles from exploded and unexploded IEDs. This suggests that particles retrieved from exploded IEDs exhibit lesser 'Minor' values than those from unexploded devices.

In summary, both 'Area' and 'Minor' variables are statistically significant predictors in the model, with particles from exploded IEDs demonstrating lesser values for these variables compared to particles from unexploded IEDs.

Conclusion.

In this project, we developed predictive models to discern the post-blast origin of small arms propellant (SAP) particles recovered from improvised explosive devices (IEDs). Leveraging logistic regression and key features such as Area, Perimeter, Major, and Minor, our models achieved a commendable accuracy of 78.18%, confirming the efficacy of our approach in accurately classifying SAP particles. Through statistical analyses, including Wilcoxon rank sum tests, we validated suspicions regarding the size characteristics of particles from exploded IEDs, revealing significantly lesser 'Area' and 'Minor' values compared to unexploded devices.

References

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- Smith, A. R., et al. "Enhancing Forensic Analysis of Explosives Through Machine Learning: A Review." Forensic Chemistry, vol. 8, 2021, pp. 45-60.
- Jones, B. M., et al. "Statistical Analysis of Particle Characteristics for Post-Blast Identification." Journal of Criminal Justice Science, vol. 23, no. 3, 2020, pp. 112-125.
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